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JOB ANALYSIS AND JOB DESIGN

INTRODUCTION

Manpower planning is concerned with determination of quantitative and qualitative requirements of manpower for the organisation. Determination of manpower requirements is one of the most important problems in manpower planning.

The quantitative aspect deals with determining the total number of people required to do the work. The kind or quality of people required to do the work comes under the qualitative aspect. Detailed knowledge of the nature and requirements of jobs to be filled is essential for determining the kind of personnel required. Such knowledge can be obtained through the process of job analysis.

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DEFINITIONS OF JOB ANALYSIS

Some of the definitions of job analysis are given as follows, to understand the meaning of the term more clearly :

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According to Michael J. Jucius, "Job analysis refers to the process of studying the operations, duties and organisational aspects of jobs in order to derive specifications or as they are called by some job descriptions."

According to Edwin B. Flippo, "Job analysis is the process of studying and collecting information relating to the operations and responsibilities of a specific job."

According to John A. Shbim "Job analysis is the methodical compilation and study of work data in order to define and characterise each occupation in such a manner as to distinguish it from all others."

According to Richard Henderson "Job analysis involves the identification and precisely identifying the required tasks, the knowledge and skills necessary for performing them and the conditions under which they must be performed."

According to S.P. Robbins & D.A. De Cenzo "Job analysis is the systematic exploration of activities within a job. It is a basic technical procedure. One that is used to define the duties, responsibilities and accountabilities of a job."

According to Herbert G Hereman "A job is a collection of tasks that can be performed by a single employee to contribute to the production of some product or service provided by the organisation. Each job has certain ability requirements (as well as certain rewards) associated with it. Job analysis is the process used to identify these requirements."

The study of these definitions reveals that job analysis is a process by which job, duties and responsibilities are defined and the information of various factors relating to jobs are collected and compiled to determine the work conditions, nature of work, qualities of persons to be employed on job, position of the job, opportunities available and authorities and privileges to be given on the job etc.

A job can be analysed only after it has been designed and someone is already performing it.

The main purpose of job analysis is thus to collect the data and then to analyse the data relating to a job. A job can be analysed only after it has been designed and someone is already performing it. Job analysis is thus performed upon ongoing jobs. As jobs are subject to change a job analysis may become obsolete within a short period of time. The data relating to the job may be classified as under :

(*i*) Job identification
(*ii*) Nature of the job
(*iii*) Operations involved in doing the job
(*iv*) Materials and equipment required to do the job
(*v*) Personnel qualities required to do the job
(*vi*) Relation of the job with other jobs in the organisation

Job description and job specification are two products of job analysis.

A careful examination of the data collected and classified suggests that some data relates to the job and some concern the individuals doing the job. The requirements of job are known as job description and the qualities required from the job holder are termed as job specifications. Thus, job description and job specification are two products of job analysis.

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OBJECTIVES OF JOB ANALYSIS

The main objectives of job analysis are as follows :

1. Work simplification

A job may be analysed to simplify the process and methods involved in it. This will mean redesigning the job. Work simplification helps to improve productivity of personnel.

2. Establishment of standards of performance

In order to hire the personnel on the scientific basis, it is very necessary to determine in advance a standard of performance with which applicants can be compared. This standard should establish the minimum acceptable qualities necessary for the effective performance of the job duties and responsibilities. These standards would be established with the help of job analysis only.

3. Support to other personnel activities

Job analysis provides support to other personnel activities such as recruitment, selection, training, development, performance appraisal, job evaluation etc.

USES OF JOB ANALYSIS / Benefits

A comprehensive job analaysis is an essential ingredient in designing a sound human resource programme. Job information gathered from job analysis process and its resulting products—job description and job specification-may be used for the following purposes :

1. **Organisational Design.** Job analysis will be useful in classifying the jobs and the interrelationships among the jobs. On the basis of information obtained through job analysis, sound decisions regarding hierarchical positions and functional differentiation can be taken. This will improve operational efficiency.

A comprehensive job analysis is an essential ingredient in designing a sound human resource programme.

2. **Manpower Planning.** Job analysis is the qualitative aspect of manpower requirements because it determines the demands of the job in terms of responsibilities and duties and then translate these demands in terms of skills, qualities and other human attributes. It also determines the quantum of work which an average person can perform on the job in a day. It facilitates the division of work into different jobs. Thus, it is an essential element of manpower planning because it matches jobs with men.

3. **Recruitment and Selection.** In order to hire a right person for a job, it is very essential to know the requirements of the job and the qualities of the individual who will perform the job. The information regarding these two are procured from job description and job specifications respectively and help the management in matching as closely as possible the job requirements with workers' aptitudes, abilities, interests etc to facilitate the execution of employment programmes.

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4. Placement and Orientation. Job analysis helps in matching the job requirements with the abilities, interests and aptitudes of people. Jobs will be assigned to persons on the basis of suitability for the job. The orientation programme will help the employees in learning the activities and understanding duties that are required to perform a given job more effectively.

5. Training and Developoment. Job analysis provides valuable information which is required to identify the training needs, to design training progammes and to evaluate training effectiveness. Job analysis helps in deciding what is to be learnt and how it is to be learnt. Employee development programmes such as job rotation, job enlargement and job enrichment are also based on the analysis of job requirements.

6. Job Evaluation. Job analysis provides a basis for job evaluation. Job evaluation aims at determining the relative worth of the jobs which in turn helps in determining the remuneration for the jobs. An accurate and comprehensive set of job descriptions and job specifications forms a factual basis for evaluating the worth of the job.

7. Performance Appraisal. Job analysis data provide a clearcut standard of performance for every job. The performance of employee can be appraised objectively with the standard of job performance so established. A supervisor can very easily compare the contribution of each man with the set standards.

8. Job Designing. Job analysis helps the industrial engineers in designing the job by making comprehensive study of the job elements. It also helps in time and motion study, work specifications, methods and work place improvement and work measurement. Human engineering activities such as physical, mental and psychological are also studied with the help of job analysis information.

9. Safety and Health. The job analysis process uncovers the hazardous and unhealthy environmental factors such as heat, noise, fumes, dust etc. The management can take corrective measures to minimise the chances of various risks to ensure safety of workers and to avoid unhealthy conditions.

10. Discipline. Job analysis studies the failure of the workers to meet the required standard of performance. Corrective measures may be taken in time to avoid untoward situations. In this way, it helps in maintaining discipline in the organisation.

11. Employee Counselling. Job analysis provides information about career choices and personal limitations. Such information is helpful in vocational guidance and rehabilitation counselling. Employees who are unable to cope with the hazards and demands of given jobs may be advised to opt for subsidiary jobs or to seek premature retirement.

12. Labour Relations. Job analysis will be helpful in improving labour management relations. It can also be used to resolve disputes and grievances relating to work load, work procedures etc.

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THE PROCESS OF JOB ANALYSIS

The process of job analysis is nothing but a data collection process. The main steps involved in job analysis are as follows :

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1. **Organising and planning for the programme.** The first step in the process of job analysis is to organise and to plan for the programme. The company must decide who will be incharge of the programme and must assign responsibilities to the designated persons. A time schedule and a budget for carrying out the analysis of job should also be prepared.

2. **Obtaining current job design information.** The job analyst should obtain information concerning the current design of the representative job. For this purpose, current job description and job specification, procedure manual, systems flow charts etc should be studied.

3. **Conducting "needs research".** The job analyst should investigate to determine which organisation, managers or staff people require job analysis or output from job analysis. The analyst should also determine for what purpose and to what extent jobs must be analysed and how the information will be used.

4. **Establishing priorities in the jobs to be analysed.** The human resource department working with various operational executives should identify the jobs to be analysed and the priority of each job analysis.

5. **Collecting job data.** The next step is to collect data about the selected jobs as they are currently being performed using established systematic techniques.

6. **Redesigning the job.** The next step is to redesign the job, if necessary.

7. **Preparing job descriptions and job classifications.** Job information which has been collected must be processed to prepare the job description form. It is a statement showing full details of the activities of the job. Separate job description forms may be used for various activities in the job and may be compiled later on. The job analysis is made with the help of these description forms. These forms may be used as reference for the future.

8. **Developing job specifications.** Job specifications are also prepared on the basis of information collected. It is a statement of minimum acceptable qualities of the person to be placed on the job. It specifies the standard by which the qualities of the person are measured. Job analyst prepares such statement taking into consideration the skills required in performing the job properly. Such statement is used in selecting a person matching with the job.

TECHNIQUES OF JOB ANALYSIS

Job analysis is mainly and essentially a data collection process. The information can be obtained in one of the following ways.

A detailed questionnaire is prepared by the job analyst and used for collecting information from the workers.

1. Questionnaire

In this method a detailed questionnaire is prepared by the Job analyst and distributed among the workers. The workers answer the questions to the best of

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their knowledge and belief. Employees' cooperation can more easily be achieved by explaining the entire programme fully in detail to the representatives of employees. Assistance of supervisors can also be sought.

This technique is *time consuming* and generally *does not yield satisfactory results* because many employees do not complete the questionnaire or furnish incorrect information because of their own limitations. The use of questionnaire is recommended only in case of those technical jobs where the job contents are not completely known to the supervisor or the operation is too complex to observe.

2. Observation Method

In the observation method, information is collected through observation of the work performed alongwith discussion with the supervisor of the job.

The most practical and reliable technique of getting information regarding job is through personal observation of the work performed, coupled with the discussion with the supervisor of the job. The analyst may ask questions from the workers on a job.

The complete and useful information can be gathered by the use of this technique. This technique can be used if a particular job is simple and of repetitive nature. Observations coupled with discussions with the supervisor and the workers, constitutes the preferred approach in most of the cases, to get the required information.

3. Interview

The job analyst contacts the workers and the supervisor concerned and asks questions regarding various jobs performed by them for collecting information.

Job analyst uses interview as a prime method of data collection. He contacts the workers and the supervisor concerned and asks questions regarding the various jobs performed by them for collecting the information. The supervisor may be taken into confidence in collecting the information about the work which the worker is not willing to supply due to certain limitations. Interview may be used as a technique for comparing the data, collected by the analyst through observation or questionnaire.

This technique can do much to dispel any doubts and suspicions which might be held. It may also be useful to crosscheck the information obtained with the incumbents of identical or similar jobs. But it may be kept in view that when information is gathered by interviewing, it is the study of positions and not of persons holding the positions.

4. Record

The analyst can also collect information from the record maintained by the personnel department.

Personnel department maintains the record of the facts regarding job and the job holders. The analyst collects the information from the record maintained by the personnel department. This method is also not free from defects. Certain information such as supervisor worker relationship, tools and accessories used and work conditions are not made available from records and hence complete information about work and worker cannot be obtained by this technique.

5. Critical Incidents

In this method, job holders are asked to describe incidents concerning the job on the basis of their past experience. The incidents so collected are analysed and classified according to the job areas they describe. A fairly clear picture of

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actual job requirements can be obtained by distinguishing between effective and ineffective behaviour of workers on the job. However, this method is time consuming. The analyst requires a high degree of skill to analyse the contents of descriptions given by workers.

A fairly clear picture of the actual job requirements can be obtained by the analysis of critical incidents.

The above techniques of collecting information are generally used but none of them is free from defects. The analyst should use these techniques in combination and not in isolation to get better results.

6. Job Performance

In this method, the job analyst actually performs the job under study to obtain a first hand experience of the actual tastes, physical and social demands and the environment of the job. This method can be used only for jobs where skill requirements are low and can therefore, be learnt quickly and easily. It is not suitable for jobs that are hazardous or for jobs that require extensive training. Moreover, it is a very time consuming method.

The job analyst actually performs the job under study to obtain a first hand experience.

Although most employers use the above methods for collecting job analysis data, there are many times when these narrative approaches are not appropriate. In that case, a more quantitative job analysis approach may be best. Following are the popular quantitative methods :

Quantitative Methods

1. Position Analysis Questionnaire. The Position Analysis Questionnaire (PAQ) is a very structured job analysis questionnaire. The PAQ itself is filled in by a job analyst, a person who should already be acquainted with the particular job to be analysed. The PAQ contains 194 items, each of which represents a basic element that may or may not play an important role in the job. The job analyst decides whether each item plays a role on the job and, if so, to that extent. 194 job elements covered by PAQ are grouped into Six Sections.

PAQ is a very structured job analysis questionnaire filled by a job analyst himself.

(*i*) **Information Input.** Where and how do workers get information to do their jobs ? 35 elements

(*ii*) **Mental Processes.** What reasoning, planning, organising and decision making is done ? 15 elements

(*iii*) **Work Output.** What physical activities, tools and machines are used ? 49 elements

(*iv*) **Relationships.** What contact with other people, both in the company and outside of it is maintained or developed ? 36 elements

(*v*) **Job Context.** What is the physical and Social Context in which the job is performed ? 19 elements

(*vi*) **Other Job Characteristics.** What other activities, conditions or characteristics not covered by other categories are relevant ? 40 elements

The advantage of PAQ is that it provides a quantitative score or profile of any job in terms of how that job rates on the basic activities. The PAQ's real

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strength is, thus, in classifying jobs. PAQ's results can be used to compare the jobs relative to one another and pay levels can be assigned for each job.

The major problem with PAQ is the time it takes for a job analyst to fill out the ratings. However, PAQ has been widely researched and tested and appears to be both reliable and valid.

2. Management Position Description Questionnaire (MPDQ). MPDQ is a standardised instrument designed specifically for use in analysing managerial jobs. The 274 item questionnaire contains 15 sections. It would take 2 hours to complete the questionnaire. In most cases the respondents are asked to state how important each item is to the position.

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W.W. Tomov and P.R. Pinto have developed the following Management Position Description factors :

(*i*) Product, marketing and financial strategy planning
(*ii*) Coordination of other organisational units and personnel
(*iii*) Internal Business Control
(*iv*) Products and services responsibility
(*v*) Public and customer relations
(*vi*) Advanced consulting
(*vii*) Autonomy of actions
(*viii*) Approval of financial commitments
(*ix*) Staff Service
(*x*) Supervision
(*xi*) Complexity and stress
(*xii*) Advanced financial responsibility
(*xii*) Broad Personnel responsibility

3. Functional Job Analysis. FJA is a worker oriented job analysis approach that attempts to describe the whole person on the job. It tries to examine the fundamental components of "data, people and things".

Functional job analysis tries to examine the fundamental components of 'data, people and things'.

There are four other dimensions which are rated under this method :

(*i*) The extent to which specific instructions are necessary to perform the task
(*ii*) The extent to which reasoning and judgement are required to perform the task
(*iii*) The mathematical ability required to perform the task and
(*iv*) The verbal and language facilities required to perform the task.

Functional Job Analysis then identifies Performance Standards and training requirements. FJA, therefore, lets you answers the question "To do this task and meet these standards, what training does the worker require ?"

FJA is frequently used for government jobs. It provides a quantitative score of each job as a function of its complexity in relationship with people, data and things. The results are helpful in fixing wage rates and in developing employee succession plans. On the negative side, FJA takes a lot of time. Training in its use may mean considerable amount of money.

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Which Method to Follow ?

There are many ways to obtain the job analysis information. The information may be obtained from individual workers, or from groups, from supervisors or from observations of the job analyst. Experts agree that the choice of a job analysis method depends upon the purposes to be served by the data. There is no best way to conduct a job analysis. Wherever possible, multiple methods of job analysis may be used e.g. a quantitative approach should be supported by a qualitative approach.

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PROBLEMS IN JOB ANALYSIS

Certain problems may crop up while conducting job analysis. The major ones are as follows :

1. Lack of Support from Top Management. In most cases the support from top management is missing. They refuse to appropriately describe what an employee is supposed to do in the company creating confusion in the minds of employees. The top management should make it clear to all employees that their full and honest participation is extremely important for the process.

2. Single Method. All too often, job analyst relies on only one of the methods when a combination of two or more methods might provide a better idea.

3. Lack of Training/Motivation. Job holders are a great source of information about the job, but they are not trained or motivated to generate quality data for job analysis. Further, job holders are rarely made aware of the importance of the data and are never rewarded for providing accurate data.

4. Distortion of Activities. When training or preparedness does not exist, job holders tend to submit distorted data, either intentionally or inadvertently. Another reason for the negative attitude is the feeling that "as long as someone does not know precisely what I am supposed to be doing, I am safe."

JOB DESCRIPTION

Job description is the immediate product of job analysis process. The data collected through job analysis provides a basis for job descriptions and job specifications. Job description functionally describes what the job entails. It discloses what, how and why the job is to be done. It classifies the job into various subjobs and analyse the nature of each subjob. The main object of job description is to differentiate it from other jobs and to set out its outer limits. Job description is a very important document as it helps to identify the job and gives a clear idea of what the job is.

Job Description discloses what, how and why the job is to be done.

DEFINITIONS

A job description is an organised factual statement of job contents in the form of duties and responsibilities of a specific job. A few important definitions of job description are as follows :

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According to Edwin B. Flippo, "The first and immediate product of job analysis is job description. As its title indicates, this document is basically descriptive in nature and constitutes a record of existing and pertinent job facts."

According to Pigors and Myres, "Job description is a pertinent picture (in writing) of the organisational relationships, responsibilities and specific duties that constitute a given job or position. It defines a scope of responsibility and continuing work assignment that are sufficiently different from that of other jobs to warrant a specific title."

The preparation of job description is necessary before a vacancy is advertised. It tells in brief the nature and type of a job.

CONTENTS OF JOB DESCRIPTION

The contents of job description are as discussed below :—

1. Proper Job Title

It is desirable that the job title should be short, definite and suggestive of the nature of the job. The desirable qualities of job holder should also be clear from the title so that every job could be distinguished from one another. Commerce and industry nomenclature (Vocabulary of names) where ever applicable should always be considered in phrasing meaningful job titles.

2. Job Summary

Job summary describes the contents of the jobs in terms of activities or tasks performed. Job summary should clear the nature of the job. Primary, secondary and other duties to be performed on the job should clearly be indicated separately.

3. Job Location

Job location should also be given in the description of the job. Job location means the place where the job is to be performed *i.e.* in which department.

4. Duties and Responsibilities

This is the most important phase of job description and should be prepared very carefully. It describes the duties to be performed along with frequency of each major duty. Responsibilities concerning custody of money, supervision and training of staff etc are also described in this part.

5. Machines, Tools and Materials

The machines, tools, equipment and material required in the performance of each job should also be included in job description. It will indicate the nature and complexity of the job and will help in devising the training programmes.

6. Working Conditions

The working environment in terms of heat, light, noise levels, dust and fumes etc, the job hazards and possibility of their occurrence and working conditions should also be described. It will be helpful in job evaluation.

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7. Nature of Supervision

The nature of supervision required on each job should also be mentioned. There are certain jobs like unskilled jobs which require close supervision while other jobs like skilled or supervisory or managerial jobs require less supervision.

8. Relation to Other Jobs

The jobs immediately below and above are mentioned. It provides an idea of vertical work flow and channels of promotion. It also indicates to whom the job holder will report and who will report to him.

PREPARATION FOR JOB DESCRIPTION

Job description provides the data of the job itself in terms of duties and responsibilities. In order to prepare the documents, the detailed information is collected about job title, job summary, duties and responsibilities etc, through any of the techniques of job analysis and compiled in a well designed bank in an organised manner. Separate blanks are filled for each job. It may be used in evaluating the job and determining the wage structure. It is also a record for future reference and guidance.

CHARACTERISTICS OF A GOOD JOB DESCRIPTION

A good job description must possess the following characteristics :—

1. It should be kept uptodate. Necessary adjustments and amendments should be made from time to time.
2. Job title should be short, definite and suggestive so as to indicate the nature of the work.
3. It should describe in sufficient detail each of the duties and responsibilities.
4. Job specifications should clearly be mentioned.
5. It should be descriptive but short and concise.
6. Statement of opinion should be avoided.
7. Examples of work performed may be quoted.
8. It should ensure that a new employee can understand the job by reading the description.
9. All employees must know the contents of job description.

In order to make it uptodate, a regular survey should be made in the organisation and necessary changes should be incorporated in the existing job description.

JOB SPECIFICATION

Job specification is also the product of job analysis. Job specification or employee specification is a statement of the minimum acceptable human qualities required for the proper performance of a job. It specifies the type of

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Job specification is a statement of the minimum acceptable human qualities required for the proper performance of a job.

person required in terms of educational qualifications, experience, aptitude etc on the job and further assists in the selection of appropriate personnel by outlining the particulars working conditions to be encountered on the job. Job specification has been defined by various authors as follows :

According to Edwin B. Flippo, "A job specification is a statement of minimum acceptable human qualities necessary to perform a job properly".

According to Dale Yoder, "The Job specification, as such a summary properly described is thus, a specialised job description, emphasising personnel requirement and designed especially to facilitate selection and placement."

It is clear from the above definitions that job specification is a statement of summary of personnel requirements for a job. It may also be called "standard of personnel for the selection"

A job specification should include :

(*i*) **Physical Characterestics** such as height, weight, chest, vision, hearing, health, age, voice, poise etc.

(*ii*) **Psychological and Social characteristics** such as emotional stability, flexibility, decision making ability, analytical view, mental ability, pleasing manners, initiative, conversational ability etc.

(*iii*) **Mental Characteristics** such as general intelligence, memory, judgement, ability to concentrate, foresight etc.

(*iv*) **Personal Characteristics** such as sex, education, family background, job experience, hobbies, extra curricular activities etc.

All these characteristics must be classified into three categories :

(a) Essential attributes which a person must possess.

(b) Desirable attributes which a person ought to posses.

(c) Contra indicators which will become a handicap to successful job perfomance.

JOB DESCRIPTION AND JOB SPECIFICATION

Both job description and job specification are the products of job analysis but these should not be confused. Following are the points of distinction between these two :—

1. Job description is the summary of the duties, responsibilities and other features of the job whereas job specification is a statement of qualities of the person who handles the job. Job specification measures the person handling the job and job description measures the job. Both these are measuring rods.
2. Job specification assists the management in selecting a match for the job. The applicants are assessed on the basis of qualities mentioned in job specification statement. The job description assists the candidate in understanding the requirements of the job which are to be fulfilled by him. It also helps in appraisal of the actual performance of the employee with job description.

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